Marginally Useful:

Formalizing the Information Gap in Conformal Prediction

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Abstract

Conformal prediction has been touted as a more formal, rigorous approach to adding uncertainty to a forecast. The sole objective of this note is to point out that rigor cuts both ways in the case of residual pooling, the technique used in the vast majority of conformal prediction applications. The fact that unconditional guarantee of coverage is provided is not in question, but we make clear, we believe for the first time, that there is an opposing guarantee too: a permanent gambit of logarithmic-score regret which no amount of data or tuning can subsequently reduce. We provide several interpretations including a financial one: the growth rate of an oracle adversary betting against odds set by someone using residual pooling. No similar self-handicap comes to mind with other statistical paradigms, and we conclude that any claim of a "rigorous advantage" is marketing, not statistics.

Keywords: Conformal prediction; proper scoring rules; calibration and sharpness; mutual information; predictive distributions.

1 Introduction

In recent years conformal prediction has gained attention in the machine learning community in particular. However, there is a danger this emphasis becomes a substitute for careful statistical modeling. In particular the use of split conformal prediction, where the field started, is essentially synonymous with the statistical advice that says one's model residuals are best modeled by their

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past empirical distribution. Such advice is arguably naive in fields like finance where stochastic volatility is a mainstay but on the other hand, there are many well-used algorithms in classification, to pick one area, where the off the shelf outputs ("proba") tend to be very poorly calibrated. In those settings, adding a conformal wrapper has real utility.

Our point is not to persuade anyone against pragmatism and we see the increased use of conformal prediction as a good sign that model residuals are getting more attention than they perhaps received from the broad data science community in the past. However it is also important, in our view, to approach any method of modeling residuals with a critical eye and the same methodological rigor that would be applied to any other statistical model. It is self-evident, in our view, that there can be no free lunch. There is nothing sacred about the empirical distribution however practical it may prove as a first remediation.

The point of this note is to convert that common sense into a crisp mathematical statement, and to suggest a working way to think about sharpness and coverage as a two-coordinate diagnostic. The kind of scenario we wish to help avoid is becoming more frequent. A practitioner reads about conformal prediction and its "rigor", tries to use it, and then is disappointed. This disappointment is sometimes manifest in a second metric such as log-likelihood that is, quite likely, more closely related to the economics at hand. The solution is always the same, careful modeling that usually will involve unconditional modeling of the residuals. It would clearly be better if the tradeoff was clear from the beginning, and that is precisely what we formalize here without trying to relitigate the well appreciated differences between certifying the coverage of a set and estimating a distribution.

It should go without saying that the split conformal guarantee is not in question. Given exchangeable data, a fitted predictor, and a nonconformity score, split conformal prediction returns a set $C_\alpha(x)$ with

$$\mathbb{P}\big(Y_{n+1} \in C_\alpha(X_{n+1})\big) \;\geq\; 1 - \alpha, \tag{1}$$

finite-sample and with no assumption on the data distribution (Vovk et al., 2005; Lei et al., 2018; Angelopoulos and Bates, 2023). The danger lies solely in the *interpretation* of (1) when a user wants a conditional distributional prediction of the next outcome. The averaging here takes place over both X_{n+1} and Y_{n+1}, and that is not at all the usual notion of prediction. The paper Lei and Wasserman (2014) is one of many warnings about this in the literature, where it is stated that "a good estimator must satisfy something more than marginal coverage".

Contribution. We will briefly review the more formal caveats to split conformal prediction and compare them to our simple exact definition of the conformal information gap. We view the conformal predictive system of Vovk et al. (2019) as an attempt to fix the residuals subject to the restriction of using only one unconditional shared shape (as compared to, for example, a data-dependent shift such as one would see in a GARCH model, say).

Once this shape is fixed, it is an identity that the expected log-score regret of that distribution relative to the conditional oracle is *exactly* the mutual information $I(R; X)$ between the residual R and the input X (Proposition 3). The reason for calling this lower bound a gap is that the act of "conformalization" itself cannot cross it, no matter how much data subsequently becomes available. It is a structural gambit[1] and common sense suggests that this gambit might be more likely to pay off if the existing base model is very poor, and less likely to pay dividends otherwise. Setting any opinions aside on that matter, it behoves the modeler to be aware of what they are sacrificing, just as it is surely informative to know the engine evaluation when playing an unsound gambit, even one with good practical results. Ultimately, modeling residuals is like modeling anything else, and reducing the gap will typically require heteroscedastic scale, quantiles, shape, multiple time scale effects and so forth.

The conformal information gap identity admits five equivalent readings (Section 5): a false-pooling cost, an average log Bayes factor, the conditional non-uniformity of conformal ranks, an information projection, and a Kelly betting rent.

Limitations of split conformal prediction glossed over Split conformal prediction and its finite-sample marginal validity are due to Vovk et al. (2005) and Lei et al. (2018); see Angelopoulos and Bates (2023) and Angelopoulos et al. (2024a) for modern treatments. The conformal predictive system (Vovk et al., 2019) is the construction that turns a conformal predictor into a full predictive distribution by re-leveling the marginal residual law.

Distribution-free conditional coverage cannot be achieved if intervals are finite length. This result appears in Lei and Wasserman (2014) and it was refined in Foygel Barber et al. (2021). In

[1] The Holloway Gambit, we might term it.

Vovk (2012) the conditional validity for inductive conformal predictors was studied. Approaches that approach the conditional objective under extra structure include Gibbs et al. (2025) and Laplante (2026).

There are numerous papers that quantify what we call the conditional gap, notably Braun et al. (2025). These necessarily condition on X or assume structure.

Despite the formal caveats, to which we add another, the applied and survey writing routinely calls the marginal-coverage guarantee "well-calibrated" or "reliable" uncertainty quantification (Bellotti and Zhao, 2025), and tutorials present conformal prediction as the way to "add uncertainty" (tds) or to "predict full probability distributions" (Manokhin).

We hasten to add that the methodological literature is more careful, separating *validity* from *efficiency*. A recent critique strand makes related points independently. Sets can be uninformative, as noted in Recht (2024). The gaming of interval length is a topic in Min et al. (2026). The misreading of conformal guarantees in applied domains is remarked on in Mehrtens et al. (2025). Our observation is morally related but not the same as the impossibility of distribution-free conditional coverage due to Lei and Wasserman (2014) and Foygel Barber et al. (2021). It is closest in spirit to the decomposition of a proper score into calibration and sharpness, which of course is standard (Gneiting et al., 2007; Gneiting and Raftery, 2007).

The calibration/sharpness decomposition and the maximize-sharpness-subject-to-calibration program are due to Gneiting et al. (2007); Gneiting and Raftery (2007), with antecedents in the prequential view of Dawid (1984). Recalibration methods (Platt, 1999; Zadrozny and Elkan, 2002; Guo et al., 2017; Kuleshov et al., 2018) adjust a forecast's probabilities.

Closest to us is Correia et al. (2024). In that work the conformal set size is related to information-theoretic uncertainty via bounds on expected set size in terms of the conditional entropy $H(Y \mid X)$. Our identity is for the log-score regret of the predictive distribution rather than for set size but the parallel should be clear. These are complementary results.

Also worthy of mention is the general trend to perform conditional prediction while marching under the banner of conformal prediction. For example conditional and shape-adaptive conformal constructions leave the single-shape class and our result does not apply to them. Examples include conformalized quantile regression (Romano et al., 2019), Mondrian/binned conformal predictive systems (Boström et al., 2021; Toccaceli, 2026), and conformal training (Stutz et al., 2022). These can buy sharpness by conditioning on X or optimizing a proper objective and trading unconditional guarantees, but so can almost every statistical approach!

Section 2 fixes notation. For context, Section 3 restates three known limits of the marginal guarantee. Section 4 reiterates that coverage and the log-score are orthogonal for a residual score. Section 5 proves the residual-information gap. Section 6 suggests a coverage score plane as a methodological crux, and the sections after it deal with time series, with the case where coverage really is the deliverable, and with the conditional constructions our result does not touch.

2 Split conformal and the marginal guarantee

Let $(X_i, Y_i)_{i=1}^{n+1}$ be exchangeable. Fix a point predictor $\hat{\mu}$ trained on a disjoint set, and a *nonconformity score* $s(x, y)$; the canonical choice in regression is the absolute residual $s(x, y) = |y - \hat{\mu}(x)|$. Compute calibration scores $s_i = s(X_i, Y_i)$ for $i = 1, \ldots, n$ and let

$$\hat{q} \;=\; s_{(k)}, \qquad k = \lceil (n+1)(1-\alpha) \rceil, \tag{2}$$

the k-th order statistic (with $\hat{q} = +\infty$ when $k > n$). The split-conformal set is

$$C_\alpha(x) = \{y : \; s(x, y) \le \hat{q}\} \;=\; [\hat{\mu}(x) - \hat{q},\; \hat{\mu}(x) + \hat{q}] \tag{3}$$

for the absolute-residual score. Exchangeability of the augmented scores makes the rank of s_{n+1} uniform, which yields (1) (Vovk et al., 2005; Lei et al., 2018). The construction is simple and the guarantee is real, and one readily verifies that empirical coverage tracks $1 - \alpha$ as the sample size, noise level, and α are varied. None of this is in dispute. The question is what it is taken to mean.

3 Known limits of the marginal guarantee

We briefly revisit the existing formalized limitations of split conformal prediction. There is a third, that exchangeability is usually absent in deployed time series, and we take that up separately in Section 7.

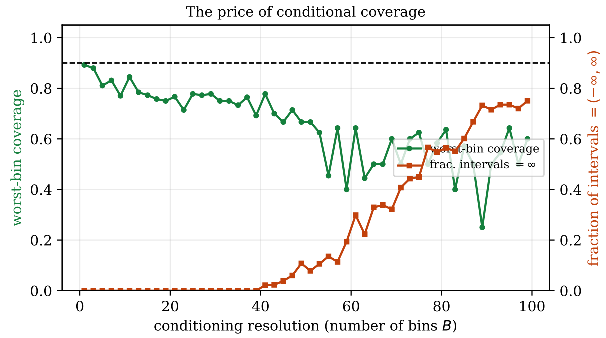

Figure 1: The price of conditional coverage (Lei and Wasserman, 2014). Per-cell calibration sets shrink and the only valid cell interval becomes $(-\infty, \infty)$.

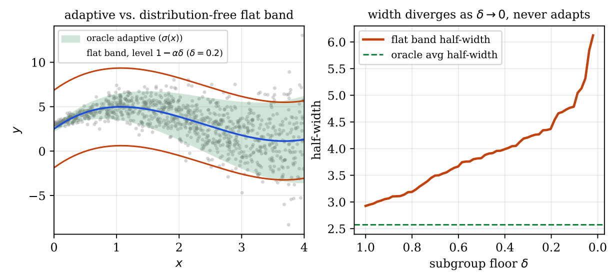

Figure 2: Subgroup coverage buys only a wider band (Foygel Barber et al., 2021). A flat, non-adaptive band at $1 - \alpha\delta$ whose width diverges as $\delta \to 0$.

(i) Marginal is not conditional, and conditional is impossible for free. In the best of all worlds one would wish for *conditional* coverage, formally $\mathbb{P}\big(Y \in C(x) \,|\, X = x\big) \geq 1 - \alpha$ for (almost) every x. However it is well appreciated that there is no way to do this without assuming something about the distribution. In Lei and Wasserman (2014) Lemma 1 it is shown that any band with non-trivial finite-sample conditional validity has *infinite* expected length at almost every point of a continuous distribution.

An even stronger case is made in Foygel Barber et al. (2021). A conditionally valid C_n has $\mathbb{E}\,\mathrm{leb}(C_n(x)) = \infty$ at almost all non-atomic x (their Prop. 2.2). Their Thm. 3.1 shows that one also cannot achieve coverage on every small subgroup (of probability at least δ) unless one inflates the coverage in a somewhat ridiculous manner, the trivial rescaling that results in totally useless bounds for any practical purpose.

The former observation we illustrate in Figure 1. The point is that as each cell's calibration count n_b is reduced until $\lceil (n_b + 1)(1 - \alpha) \rceil > n_b$ we have by (2) that $\hat{q} = +\infty$ and the only valid cell interval is $(-\infty, \infty)$.

Thus chasing uniform coverage in this manner feels like chasing a mirage. The second finer point is Figure 2. The distribution-free protection of all δ-subgroups is exactly the flat band at level $1 - \alpha\delta$. The width has to diverge as $\delta \to 0$. It does not adapt to x and the penalty is the multiplicative factor.

The Faustian bargain is clear. The guarantee you can have distribution-free is the marginal one. On the other hand a marginally valid band "tends to overestimate the set when x is in the high

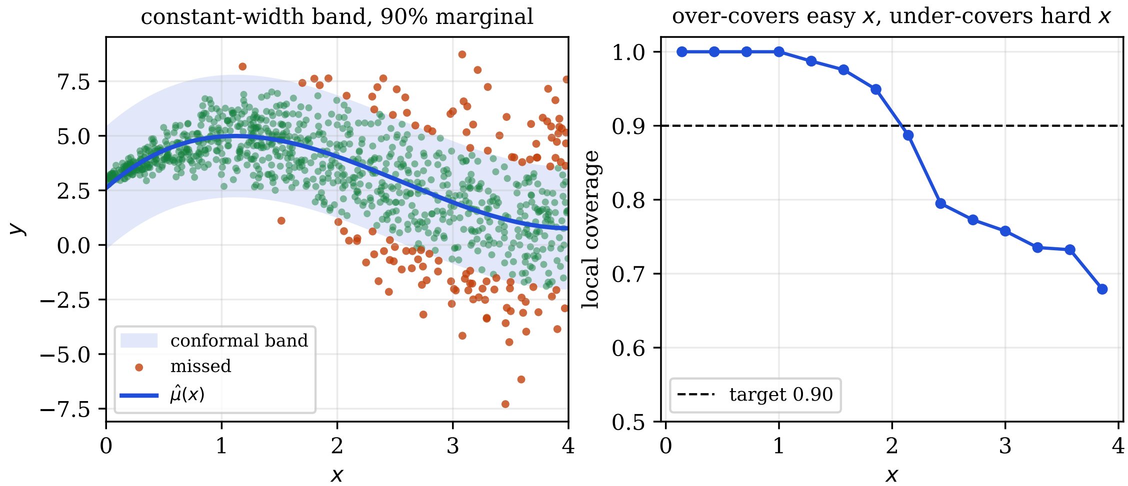

Figure 3: Marginal is not conditional. A constant-width split-conformal band (left) attains 90% *marginal* coverage on heteroscedastic data, but its *local* coverage (right) runs near 100% where the noise is small and well below target where it is large: it over-covers the easy inputs and under-covers the hard ones.

density area and to underestimate for low density x" (Lei and Wasserman, 2014). It over-covers the easy inputs and under-covers the hard ones. That's the game. To make it all the more plain, Figure 3 exhibits exactly this on heteroscedastic data: 90% overall, near-100% where the noise is small, well below target where it is large.

(ii) Validity is trivially satisfiable. The conformal literature already separates *validity* from *efficiency* and it is worth a short review of that manner of gaming also. A predictor that returns the whole outcome space with probability $1-\alpha$ and the empty set otherwise satisfies (1) exactly. But clearly this is silly. It conveys nothing. It is a noted concern that the very *same* coverage attaches to an excellent $\hat{\mu}$ and to a deliberately terrible one! All the terrible one has to do is accept a wider $\hat{q}$ in (2).

4 Coverage and score are orthogonal

Marginal validity is clearly not, on its own, evidence of a good method. Log-likelihood is a reliable companion. We would argue that they belong on separate axes, as the two separate things a forecaster reports. Those can be a point predictor $\hat{\mu}(x)$ and a full predictive density $f(\cdot \mid x)$. The log-score is a natural way to assess f (including its sharpness). The canonical residual score

$s(x,y) = |y - \hat{\mu}(x)|$ uses only $\hat{\mu}$, the *location* of the forecast. It never sees the spread or shape of f. So the conformal set, and its coverage, are invariant to exactly the part of f that the log-score rewards.

We wish to comment rigorously on the notion of "attaching uncertainty" using conformal prediction. The split-conformal set (3) is a measurable function of the nonconformity scores $\{s_i\}$ and of $s(x,\cdot)$. It is not a functional of the predictive density f unless f is explicitly used in the score, which it isn't if we are following the spirit of this common advice to "attach uncertainty". To be mildly formal:

Proposition 1 (Residual-score coverage does not constrain the log-score)**.** *Two forecasters with the* same *location $\hat{\mu}$ (hence, for the residual score, the same score function and calibration scores) but different predictive densities f have* identical *conformal sets, and identical marginal coverage at every level α, while their expected log-scores $\mathbb{E}[\log f(Y \mid X)]$ may differ by an arbitrary amount.*

Proof. The first claim is immediate from (2)–(3): $\hat{q}$ is an order statistic of $\{s_i\}$ and $C_\alpha(x) = \{y : s(x,y) \le \hat{q}\}$, so two forecasters agreeing on s and on $\{s_i\}$ produce the same set for every x, α, and (1) concerns only that set.

For the gap, take X degenerate (a point mass at an arbitrary x_0, i.e. a single repeated input, so there is nothing to vary across) and $Y \sim N(0,1)$. Both forecasters share the location $\hat{\mu} \equiv 0$, so both use the absolute-residual score $s(x,y) = |y|$, which depends on $\hat{\mu}$ but not on the spread of the predictive density. They differ only in that claimed density, $f_\sigma = N(0,\sigma^2)$: for any σ the scores are $|Y_i|$, independent of σ, so the conformal set is the fixed interval $[-\hat{q},\hat{q}]$ with $\hat{q}$ the empirical $(1-\alpha)$-quantile of $|Y|$. Yet

$$\mathbb{E}[\log f_\sigma(Y)] \;=\; -\tfrac{1}{2}\log(2\pi\sigma^2) - \frac{\mathbb{E}[Y^2]}{2\sigma^2} \;=\; -\tfrac{1}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \;\xrightarrow[\sigma\to 0^+]{}\; -\infty,$$

and likewise $\to -\infty$ as $\sigma \to \infty$. Hence for any $M > 0$ there are σ_1, σ_2 with identical conformal sets (identical marginal coverage at every α) yet log-scores differing by more than M. $\square$

Corollary 1. *For a residual-score conformal predictor, marginal validity certifies an order statistic of residual magnitudes. It does not identify, bound, or rank the scale, shape, or sharpness of any predictive density that was not used to construct the score; in particular, no inference about forecast*

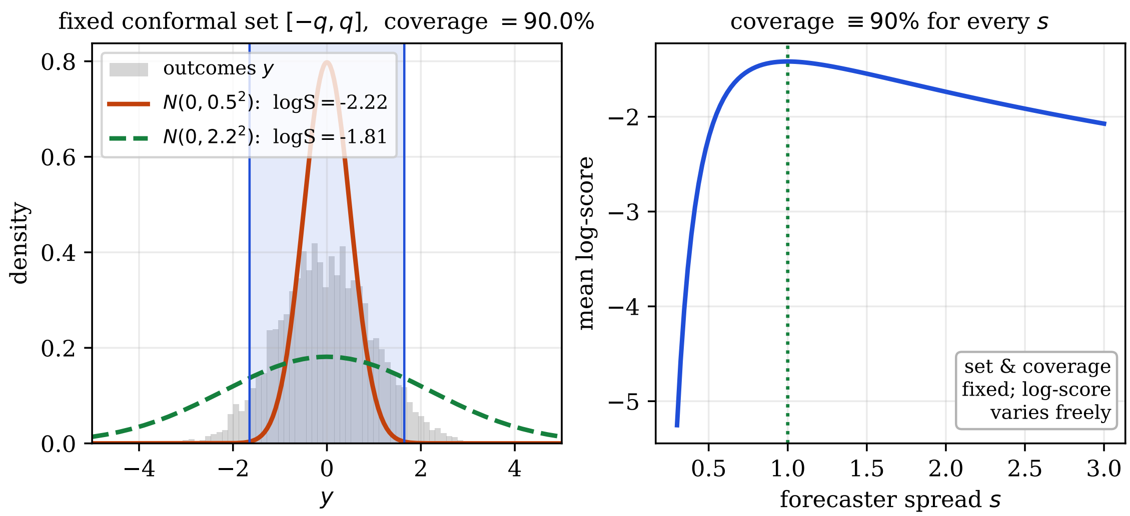

Figure 4: Coverage does not constrain the log-score (Proposition 1). With a residual score the conformal set $[-q, q]$ is fixed by the outcomes alone (left), so two predictive densities with very different log-scores share the *same* set and the same 90% coverage. Sweeping the forecaster's spread s (right) moves the log-score freely while coverage stays pinned at 90%.

quality (as graded by a strictly proper score) can be drawn from coverage alone.

Remark 1 (The degenerate X is only for cleanliness)**.** The point-mass X in the proof makes the unbounded gap transparent, but it is not needed. For any non-degenerate X, take two forecasters sharing the same location $\hat{\mu}$ (hence, for the residual score, the same score function and calibration scores, so identical sets and coverage at every α) but predictive spreads $\sigma_1(x), \sigma_2(x)$; their expected log-scores differ by $\mathbb{E}_X[\log(\sigma_2/\sigma_1)] + \frac{1}{2}\mathbb{E}_X[(\sigma_2^{-2} - \sigma_1^{-2})\,\mathrm{Var}(Y \mid X)]$, which is unbounded over choices of the σ_i. Coverage is blind to all of it.

This is nothing more than a formal version of a point numerous authors have been making informally. Perhaps the most blunt of these commentators is Recht (2024), who writes that conformal validity "almost invites you to use garbage prediction functions". Figure 4 illustrates that criticism, because the spread slides freely while the conformal interval and its 90% coverage do not move. How could it? The spread never entered (2).

5 The residual-information gap

Do we need another formal reminder of the conformal Faustian bargain? Forecast quality decomposes into *calibration* and *sharpness*, and the discipline's stated goal is to maximize sharpness subject to calibration, assessed by a proper scoring rule (Gneiting et al., 2007; Gneiting and Raftery, 2007). We point out that within the fixed-location, single-shape residual class, signed CPS is log-score optimal. It was not lost but gone before, we might say. That is why we have termed it a gambit. The conformal calibration step is doing its part, but that is simply not enough.

Conformal predictive systems. A conformal predictor can be made to output a distribution rather than a set, the *conformal predictive system* (CPS) of Vovk et al. (2019), built from *signed* residual scores. It is the mode in which conformal prediction estimates a full distribution, so it is the form the criticism must address. We must, however, distinguish it from the more common practice of reading the nested symmetric intervals of ordinary *absolute*-residual split conformal as if they were a distribution; the two re-level different objects, and the distinction matters for the gap below.

Proposition 2 (Re-leveling, two cases)**.** *Fix a location predictor $\hat{\mu}$ with residual $R = Y - \hat{\mu}(X)$ of marginal density g and CDF G. In the large-calibration limit (under exchangeability):*[2]

(A) Signed-residual CPS. The CPS predictive distribution is $\Pi_x(y) = G\big(y - \hat{\mu}(x)\big)$ (the calibration ECDF $\widehat{G}_n \to G$ by Glivenko–Cantelli), with density $\pi_x(y) = g\big(y - \hat{\mu}(x)\big)$: the base location forecast re-leveled by the marginal signed-residual law.

(B) Absolute-residual intervals. Ordinary split conformal at level α returns the symmetric interval $\hat{\mu}(x) \pm \hat{q}_\alpha$, with $\hat{q}_\alpha$ the $(1-\alpha)$-quantile of $|R|$. Read as a predictive distribution (by sweeping α) it is symmetric about $\hat{\mu}(x)$ with $|\cdot|$ distributed as $|R|$; its density is the symmetrized *marginal residual law $h_{\mathrm{sym}}(z) = \frac{1}{2}\big(g(z) + g(-z)\big)$. This is not an additional conformal guarantee; it is the density implicitly obtained if the nested absolute-residual intervals are read as central credible intervals.*

Proof sketch. (A) The CPS at level α returns endpoints equal to the signed-residual order statistics; sweeping α traces $\hat{\mu}(x)$ plus the residual quantile function, i.e. $\widehat{G}_n^{-1}$ shifted by $\hat{\mu}(x)$, so $\Pi_x(y) = \widehat{G}_n(y - \hat{\mu}(x))$; take limits (Vovk et al., 2019). (B) The intervals depend on the data only through $|R|$ and are symmetric about $\hat{\mu}(x)$; the unique symmetric density whose absolute value has the law of $|R|$ (with signed marginal g) is $h_{\mathrm{sym}}(z) = \frac{1}{2}(g(z) + g(-z))$. □

[2]The log-score comparison is for the limiting residual density, or equivalently a smoothed CPS density: a finite empirical CPS is a valid predictive distribution but not generally a Lebesgue density, so the density/log-score identity is an asymptotic (or smoothed-density) statement. The finite-sample guarantee is coverage.

In both cases the estimation was done by the base model and the residual sample; conformal supplies only the leveling. We now quantify the cost. Write the true conditional residual density as $r(\cdot\,|\,x)$, so the oracle density is $q^\star(y\,|\,x) = r\big(y - \hat\mu(x)\,|\,x\big)$, and let $\bar r = \mathbb{E}_X\, r(\cdot\,|\,X) = g$ be the marginal residual density.

Proposition 3 (Residual-information gap)**.** *Fix the location predictor $\hat\mu$ and write $I(R;X) = \mathbb{E}_X\,\mathrm{KL}\big(r(\cdot\,|\,X)\,\|\,\bar r\big) \geq 0$, the mutual information between the residual and the input. In the large-calibration limit of Proposition 2 (so that the signed-CPS shape is $\bar r$ and the absolute-residual shape is h_{sym}), and assuming the relevant densities and information quantities are finite, the expected log-score regret relative to the oracle $q^\star$ is*

$$\text{(A) signed CPS:}\quad \mathbb{E}[\log q^\star(Y\,|\,X)] - \mathbb{E}[\log \pi_X(Y)] \;=\; I(R;X), \tag{4}$$

$$\text{(B) absolute intervals:}\quad \mathbb{E}[\log q^\star(Y\,|\,X)] - \mathbb{E}[\log h_{\mathrm{sym}}(R)] \;=\; I(R;X) + \mathrm{KL}\big(\bar r\,\|\,h_{\mathrm{sym}}\big). \tag{5}$$

Both are ≥ 0. The term $I(R;X)$ vanishes iff $R \perp X$; the extra term $\mathrm{KL}(\bar r\,\|\,h_{\mathrm{sym}})$ vanishes iff the marginal residual law is symmetric. Among all single-shape forecasters $y \mapsto h(y - \hat\mu(x))$, $\mathbb{E}[\log h(R)]$ is maximized at $h = \bar r$; the signed CPS attains this optimum. Hence no recalibration that ignores X can reduce $I(R;X)$: it can at best replace the shape by $\bar r$, removing the skewness term in (5) *and reducing case (B) to case (A). Reducing $I(R;X)$ itself requires conditioning on X.*

Proof. Write H for the residual law with density h, and let $\mathcal{R}(h)$ be the expected log-score regret of the single-shape forecaster $y \mapsto h(y - \hat\mu(x))$ relative to the oracle. The oracle scores Y by $\log q^\star(Y\,|\,X) = \log r(R\,|\,X)$ and this forecaster by $\log h(R)$; the joint $P_{X,R}$ has density $p(x)\,r(\rho\,|\,x)$ and the product reference $P_X \otimes H$ has density $p(x)\,h(\rho)$, so

$$\mathcal{R}(h) = \mathbb{E}\left[\log \frac{r(R\,|\,X)}{h(R)}\right] = \mathrm{KL}\big(P_{X,R}\,\big\|\,P_X \otimes H\big) = I(R;X) + \mathrm{KL}(\bar r\,\|\,h), \tag{6}$$

the last step the chain rule for relative entropy along a product reference (the R-marginal of $P_{X,R}$ is $\bar r$, and $I(R;X) = \mathrm{KL}(P_{X,R}\,\|\,P_X P_R)$). Taking $h = \bar r$ kills the second term and gives (4); taking $h = h_{\mathrm{sym}}$ gives (5). The second term is ≥ 0 and is minimized at $h = \bar r$ (Gibbs), so no X-blind shape improves on $I(R;X)$, and $I(R;X) = \mathrm{KL}(P_{X,R}\,\|\,P_X P_R)$ is reduced only by conditioning on X. $\square$

Remark 2 (What this says, and does not say)**.** The right-hand side of (4) is exactly the information about the residual distribution that remains in X after the location predictor is fixed. Calling it the residual-information gap gives "blind to conditional residual shape" a number. Three caveats are worth stating plainly.

(1) *Conformal is not dominated within its own class.* The signed CPS is log-score optimal among single-shape location forecasters. What it pays for is the class restriction, not the calibration step. It is the absolute-residual interval system that pays the extra $\mathrm{KL}(\bar{r} \,\|\, h_{\mathrm{sym}})$, and it pays it for discarding sign information. Since $h_{\mathrm{sym}} \geq \frac{1}{2}\bar{r}$ that penalty is at most $\log 2$, one bit. The structural loss is $I(R; X)$.

(2) *What recalibration can and cannot do.* A recalibration that ignores X, marginal PIT recalibration for instance, can move the shape toward $\bar{r}$ and so erase the skewness term. That turns case (B) into case (A). It cannot touch $I(R; X)$. Reducing $I(R; X)$ means modeling conditional residual shape, whether by heteroscedastic models, conditional recalibration (Kuleshov et al., 2018), or conformalized quantile regression (Romano et al., 2019), the last being the conformal world's own way of leaving the single-shape class.

(3) *The distribution-free guarantee is real.* Recalibration offers calibration only in expectation or asymptotically and carries no coverage certificate. The conformal marginal guarantee is finite-sample and assumption-light. Where the certificate itself is the deliverable, that is decisive, and we return to it in Section 8.

Remark 3 (Five readings of the gap)**.** One number, but there is more than one way to look at it. The identity $\mathcal{R}(\bar{r}) = I(R; X) = \mathrm{KL}(P_{X,R} \,\|\, P_X P_R)$ admits at least five readings, and different readers will find different ones persuasive.

(i) A false-pooling cost. A single-shape system asserts that, once $\hat{\mu}$ is fixed, one residual law fits everyone. That is the assertion $R \perp X$. The gap is the relative entropy from the true residual experiment to the nearest such independence model. It is the log-score price of the assumption, and one pays it whether or not one notices making it.

(ii) An average log Bayes factor. Since $I(R; X) = \mathbb{E} \log \frac{r(R \mid X)}{\bar{r}(R)}$, every observation contributes the log-likelihood ratio of its own x-specific residual law against the pooled one. The gap is the oracle's expected advantage per sample.

(iii) Conditional non-uniformity of conformal ranks. Let $U = G(R)$ be the rank of the residual under the pooled law, which is the PIT used by signed CPS (assuming G is continuous, and using the randomized PIT otherwise). Marginally U is uniform, and that is the conformal achievement. But U is an invertible transform of R, so $I(R;X) = \mathbb{E}_X \operatorname{KL}\big(P_{U \mid X} \,\|\, \mathrm{Unif}[0,1]\big)$ is the *conditional* non-uniformity of those same ranks. Easy inputs concentrate U near $\frac{1}{2}$ and hard inputs push it into the tails. The single pooled shape cannot tell the two apart, as Figure 5 shows.

(iv) A projection. Equation (6) is the information projection of $P_{X,R}$ onto the single-shape product family $\{P_X \otimes H\}$. The optimizer is $H = P_R$, that is $P_X \otimes P_R$, and the projection loss is $\operatorname{KL}(P_{X,R} \,\|\, P_X P_R) = I(R;X)$. No X-blind step can cross it. The missing information is dependence, not marginal shape, and no amount of reshaping a marginal supplies dependence.

(v) A betting rent. By the log-optimal (Kelly) growth theorem, a gambler who knows a distribution P and bets against a fair book priced by Q compounds wealth at expected log-rate $\operatorname{KL}(P \,\|\, Q)$ per round (Kelly, 1956; Cover and Thomas, 2006). Read $\bar{r}$ as the book posted by a single-shape conformal forecaster and $r(\cdot \mid X)$ as what the oracle knows. The oracle's expected rent per round is $\mathbb{E}_X \operatorname{KL}\big(r(\cdot \mid X) \,\|\, \bar{r}\big) = I(R;X)$, which is reading (ii) again, now denominated in log-wealth. The conformal step fixes the level of the book, meaning its normalization and hence its coverage. It does not touch the rent. A re-leveling that ignores X leaves the divergence alone, and so leaves the growth rate alone.

Remark 4 (The gap is not an artifact of a poor location fit)**.** The quantity $I(R;X)$ is defined for a *fixed* $\hat{\mu}$, so a natural worry is that it is really just a measure of how badly $\hat{\mu}$ fits. It is not. Take the Bayes-optimal location $\hat{\mu}(x) = \mathbb{E}[Y \mid X = x]$, which zeroes the conditional mean of R. Under heteroscedasticity, or under a shape that changes with x, the *conditional law* of the residual still varies, so $I(R;X) > 0$ in general. What the gap measures is conditional residual *scale and shape*, and that is orthogonal to getting the location right. Improving $\hat{\mu}$ can raise or lower $I(R;X)$. The lever the gap points at is conditioning the residual shape on X, not refining the point forecast.

The ingredients here are individually well known. What is new is the single exact identity along the score axis. For the single-shape system the log-score regret to the oracle equals the conditional residual information $I(R;X)$, with an additional symmetrisation term when one reads absolute-residual intervals as a distribution. The quantity conformalization cannot touch now has a

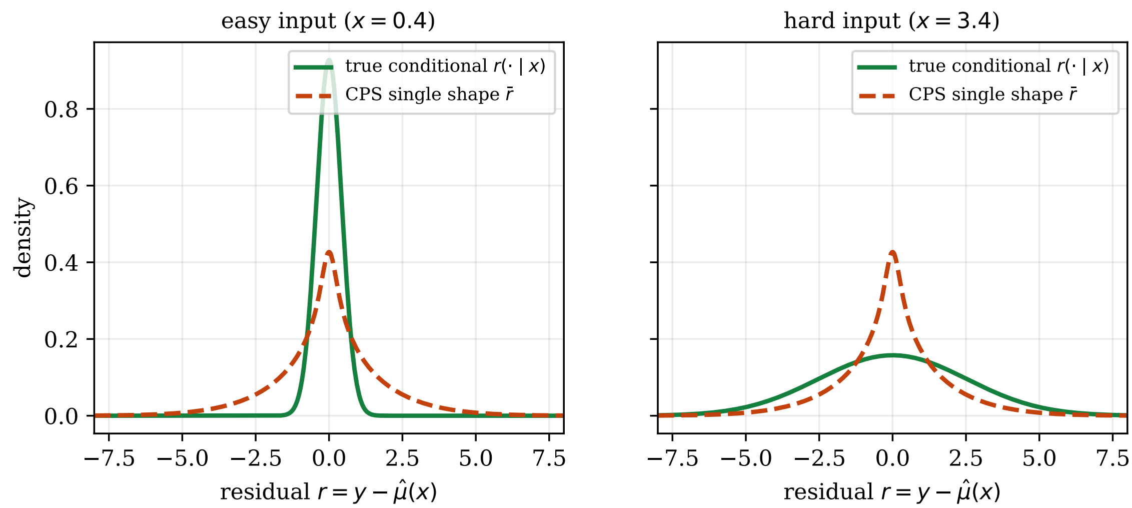

Figure 5: The residual-information gap (Propositions 2–3). A single-shape conformal predictive system uses the marginal residual law $\bar{r}$ for every input; the oracle uses the true conditional $r(\cdot \mid x)$, narrow for easy inputs (left) and wide for hard ones (right). The expected log-score regret is exactly the mutual information $I(R; X)$, which no X-ignoring recalibration can reduce.

name and a closed form, and it can be computed.

Figure 5 draws the gap. Under heteroscedasticity the true conditional residual law $r(\cdot \mid x)$ varies with x while the single-shape system uses the marginal $\bar{r}$ everywhere, and the regret is exactly the information $I(R; X)$ that separates the two. The special case $I(R; X) = 0$ is instructive. Take $\hat{\mu}$ correct and r a fixed Gaussian with the wrong scale. There, variance recalibration fitted to the log-score recovers the shape and improves the score, whereas conformalization re-levels to exact coverage and hands back a set.

6 The coverage–score plane

All of this suggests a single diagnostic, provided one is careful about what is being plotted. Define a *reporting system* as a set-valued report $C_\alpha(x)$ and, optionally, a density-valued report $f(\cdot \mid x)$. Its coordinates are

$$\big(C,\, S\big), \qquad C = \big|\,\mathbb{P}(Y \in C_\alpha(X)) - (1-\alpha)\,\big|, \qquad S = \mathbb{E}[\log f(Y \mid X)], \tag{7}$$

the marginal-coverage error of the *set* and the proper-score performance of the *density*. The calibration and sharpness decomposition (Gneiting et al., 2007) amounts, in this plane, to the statement that the two coordinates are distinct. Our results then read geometrically.

- Residual split conformal is a horizontal projection. It maps a report $(C_\alpha^{\text{base}}, f)$ to $(C_\alpha^{\text{conf}}, f)$, changing the set so that $C \to 0$ and leaving the density f, and hence S, exactly where it was (Proposition 1). It moves a point left, never up.

- Conformal predictive systems are a different matter. They supply their own density-valued report, so they do have a vertical coordinate. But that coordinate is governed by the single-shape restriction of Proposition 3, and it is capped at the oracle minus $I(R; X)$.

- Recalibration and better modeling move vertically. Fitting shape to a proper score raises S, and conditioning on x is the only way to raise it past the ceiling set by the residual-information gap (4).

A system reported by its coverage alone is a system reported by one coordinate of a two-coordinate quality. We make no great claim for the plane itself. It is a presentation device, and a close relative of reliability and sharpness diagrams (Gneiting et al., 2007). But the "conformal moves left, never up" reading makes the distinction hard to miss, which is the whole point of drawing it. It also sharpens the connection to Correia et al. (2024), whose set-size and entropy bound lives on the *efficiency* coordinate, the one that carries information, while the marginal-coverage coordinate carries none by Proposition 1. Figure 6 is the schematic. Conformalization is the horizontal move, left toward exact coverage and never up. X-blind recalibration is the vertical move, but only as far as the single-shape ceiling. The residual-information gap $I(R; X)$ between that ceiling and the oracle is crossed only by conditioning on X.

7 Exchangeability and the time-series case

Drop exchangeability and (1) is no longer guaranteed. The repairs in the literature are ingenious and we do not fault them. They are uniform in one respect, though. Each survives by redefining what is guaranteed.

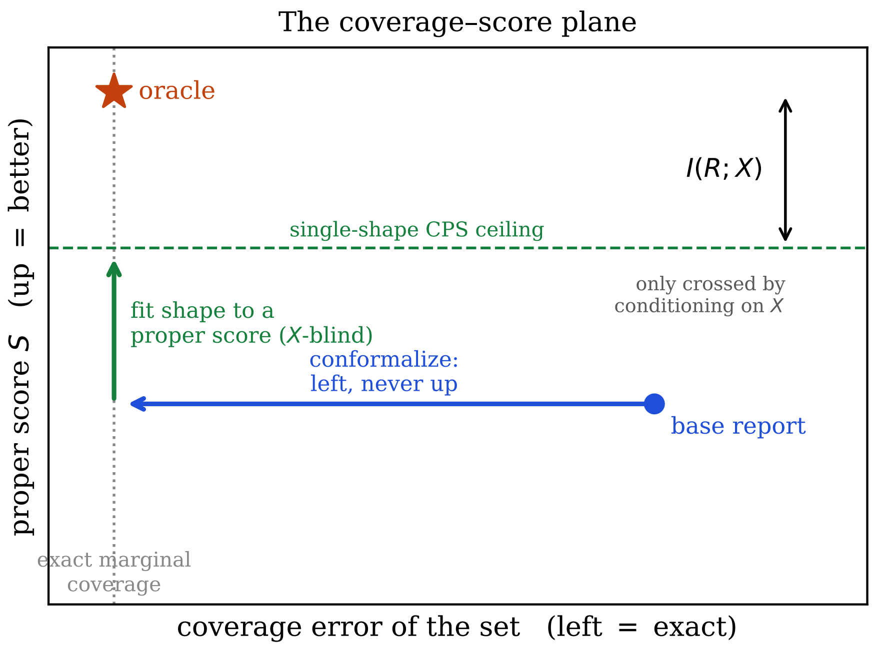

Figure 6: The coverage–score plane (schematic). The two coordinates are distinct: conformalizing a residual-score report moves it horizontally toward exact marginal coverage without changing its proper score (Proposition 1); fitting the residual shape to a proper score moves it up, but an X-blind shape can rise only to the single-shape ceiling (Proposition 3); the remaining residual-information gap $I(R;X)$ to the oracle is closed only by conditioning on X.

- Weighted conformal prediction (Tibshirani et al., 2019) restores marginal coverage under covariate shift when the likelihood ratio is known; with estimated weights coverage becomes approximate and the effective sample size drops.
- Adaptive Conformal Inference (Gibbs and Candès, 2021) updates the level online and "puts no constraints on the data generating distribution," but guarantees a *long-run time-average* miscoverage frequency, $\frac{1}{T}\sum_t \mathrm{err}_t \to \alpha$, not a finite-sample statement about $\mathbb{P}(Y \in C(X))$, and certainly not conditional coverage.
- EnbPI (Xu and Xie, 2021) drops exchangeability for approximate marginal coverage under strong-mixing/stationarity surrogates; later online methods (Zaffran et al., 2022; Gibbs and Candès, 2024) retreat to regret bounds over local intervals. The general beyond-exchangeability analysis of Barber et al. (2023) bounds the coverage gap by a total-variation distance that is exactly what nonstationarity inflates.

So three distinct guarantees now travel under the one word "conformal". Split conformal under exchangeability gives finite-sample marginal coverage. Adaptive conformal inference gives long-run, time-average coverage control under distribution shift. Online conformal under arbitrary shifts gives regret-style or local-interval performance instead of the original exchangeable guarantee. These are progressively weaker and more diffuse objects, and not one of them is the sharp predictive distribution *now*, for *this* step, that a proper score grades.

None of which takes anything away from their utility. Long-run, time-average frequency control under drift is a genuine and hard-won engineering guarantee, and it is often the most one can ask for without a model of the shift. It is simply a weaker and differently posed object than conditional coverage now. Meanwhile the exchangeability the first guarantee needs is typically unavailable in deployed time series, although blocking or learned-transform constructions can sometimes recover something like it.

Figure 7 makes the consequence concrete. On a nonstationary series, fixed split conformal collapses far below its nominal coverage once the volatility regime shifts, because exchangeability has failed. The online adaptive variants recover a long-run coverage *frequency* by re-leveling to recently realized residuals. This is exactly what the gap of Section 5 predicts. Where adaptivity helps, it helps by *local conditional modeling* of the error scale, with the conformal step supplying the coverage leveling on top. Never the sharpness. A simple probabilistic model that estimates the conditional spread directly, a streaming exponentially-weighted volatility forecaster say (Cotton, 2021), tracks the target and returns a full predictive distribution, and conformalizing it leaves any proper score essentially unchanged. That is the experience the introduction describes.

8 When coverage *is* the objective

Our argument is skeptical, not dismissive, and it comes with a precise converse. The litmus is a single question:

> *Is your loss a function of* whether *the truth lands in a region, or of* where *it lands?*

If the former, then coverage is the native object, and conformal prediction is sound and often ideal.

- Selective prediction and risk control. "Return a small set guaranteed to contain the label 95% of the time, and a human will check it." The deliverable really is a set. Risk-controlling

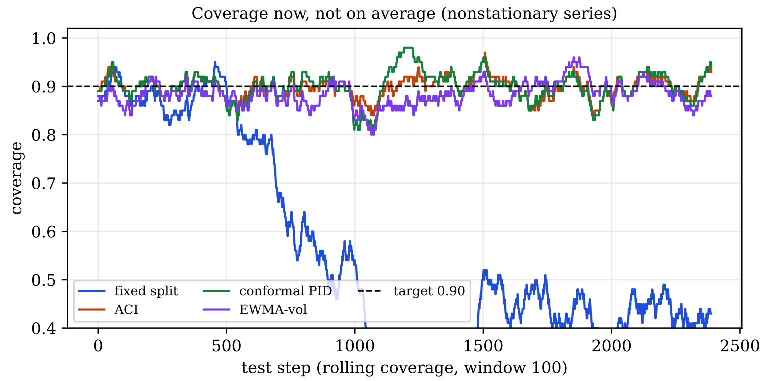

Figure 7: Coverage now, not on average. Rolling coverage on a nonstationary series: fixed split conformal collapses once the volatility regime arrives, while adaptive conformal and a probabilistic volatility model track the target. Even the recovering methods oscillate around the target rather than holding it pointwise.

prediction sets and conformal risk control formalize this (Angelopoulos et al., 2021; Bates et al., 2021; Angelopoulos et al., 2024b).

- Retrieval and shortlisting, where the product is a candidate set and coverage is recall.
- Anomaly and novelty detection via conformal p-values. This is conformal prediction's most natural home, because here it is not a forecaster at all but a distribution-free hypothesis test, and coverage is Type-I error control (Vovk et al., 2005).
- Compliance and long-run frequency control, where a certificate of the form "contained 95% of the time on average" *is* the product. The distribution-free, finite-sample nature of (1) is exactly the value added.

Two caveats temper even these. If one trusts a calibrated posterior one can form a sharper set oneself, namely the smallest region of mass $\geq 1 - \alpha$, and recover the expectations that a bare set throws away. The marginal value of conformal prediction is then purely the distribution-free guarantee, which is worth having when one distrusts the density and exchangeability holds. The second caveat is that even when coverage is the goal, one usually wants it conditionally, and that returns us to the impossibility of Section 3.

9 Scope, and the constructive escapes

Our strong claims target one object, the one that is actually over-sold. That is *post-hoc, marginal, split conformal prediction with a residual score*. A large and active literature builds conformal methods that do considerably more. The conditional constructions below, CQR and Mondrian or binned CPS, are standard practice by now rather than niche alternatives, and we do not question their effectiveness for a moment. How they relate still matters, because in every case they confirm the mechanism rather than contradict it. They buy sharpness or conditional coverage by *conditioning on X or optimizing a proper objective*, which is precisely what Proposition 3 says is required. They buy it by modeling, in other words, and not by anything the conformal step itself supplies.

Conformal predictive systems that condition on X. Mondrian or binned conformal predictive systems (Boström et al., 2021) fit a separate residual law per region, and Toccaceli (2026) chooses the bins by minimizing a leave-one-out CRPS. These produce sharp, shape-adaptive predictive distributions. They do so by leaving the single-shape class of Proposition 2, since a per-bin shape is X-dependent, and in the CRPS case by making the partition criterion a proper score. This is the most pointed case for the defence and the concession is a real one. When the reference class is itself selected by a proper scoring rule, the conformal *construction* is performing conditional density estimation. The distinction we are drawing does not vanish. It relocates, to the question of which component is estimating the distribution, and the answer is the conditional partition rather than the coverage certificate. The value sits in the estimation, as it does throughout. Conformalized quantile regression (Romano et al., 2019) is the same story with a quantile model standing in for the bins.

End-to-end conformal training. The orthogonality of Proposition 1 is a statement about *post-hoc* conformalization of a fixed predictor. Conformal training (Stutz et al., 2022; Correia et al., 2024) deliberately differentiates through the conformalizer to minimize expected set size subject to coverage, which couples the coverage and efficiency axes by construction. This does not contradict the proposition. It illustrates the corollary, which is that to obtain sharpness one must optimize for it, and cannot read it off a coverage certificate.

The information in set size. Correia et al. (2024) connect conformal set size to information-theoretic uncertainty, with bounds relating expected prediction-set size to the conditional entropy $H(Y \mid X)$. This is complementary to our Proposition 3, in that both say conformal *geometry* carries information. It also sharpens the two-coordinate plane of Section 6. The *efficiency* coordinate, set size or sharpness, is the one that bears information, while the marginal *coverage* coordinate does not, by Proposition 1. Report only the latter and one has discarded precisely the coordinate that encodes aleatoric uncertainty.

Approaching conditional coverage. The no-go result of Section 3 concerns *exact, distribution-free, finite-length* conditional coverage, and it is not the end of the subject. Diagnostics such as the excess-risk-of-target-coverage family (Braun et al., 2025) quantify and decompose the conditional gap. Methods that target conditional or subgroup-wise validity directly interpolate between marginal and conditional coverage (Gibbs et al., 2025). Post-processing via pivotal scores and optimal transport (Laplante, 2026) attains *approximate* conditional coverage by estimating the conditional law of the score, which is again conditioning on X, and explicitly under the structural assumptions the impossibility theorem rules out for the exact, assumption-free case. All of these approach the goalpost. None of them moves the one the no-go result defends.

Modern time-series conformal. Beyond the adaptive methods discussed in Section 7, methods such as HopCPT (Auer et al., 2023) exploit temporal structure to tighten intervals and improve empirical adaptivity. Their guarantees remain of the approximate or long-run kind catalogued there, and the improvement comes, once again, from modeling the conditional error law.

Beyond the log-score. We used the log-score because it gives clean information-theoretic identities. The two-coordinate distinction, coverage against score, carries over to any strictly proper score. The exact mutual-information identity does not. It is special to the log-score, and for CRPS or the interval score the analogous gap is the regret of the best X-blind residual law under that score. The accompanying benchmark in fact ranks methods by the interval (Winkler) score and CRPS (Gneiting and Raftery, 2007), and the same orderings hold there.

10 Prediction versus verification

Conformal prediction *verifies* a coverage property. It does not *model*, and a fixed rule is no substitute for modeling. Conformalization is best viewed as a terminal certification operator. It repairs the marginal coverage coordinate of a set-valued report, and any improvement in a proper score has to come from upstream modeling of the conditional location, scale, quantiles, or residual shape. It moves the coverage coordinate and nothing else. The residual-information gap is reduced by conditioning on X, which is what the conditional-shape models of Section 9 do, and never by the conformal step.

Practical remarks. Three suggestions follow, and none of them is exotic. Conformalize *last*. Model first and then add the certificate, since the conformal step cannot improve any proper score, so one should estimate first and certify second. Use a proper score, log-likelihood or CRPS, as the sharpness diagnostic. Because conformalization leaves it unchanged, a score that does not move after conformalizing is confirmation that whatever gain one has came from the modeling. And to close $I(R;X)$, add conditioning to the model, in the form of heteroscedastic spread or quantile or residual-shape estimation. Do not add tightness to the conformal step. There is nothing there to tighten.

Limitations. The exact result is deliberately narrow and it is worth saying where it stops. On *scope*, the strong claims concern post-hoc, marginal, split conformal with a residual score, and the single-shape residual class. The conditional constructions of Section 9, meaning CQR, Mondrian or binned CPS, and conformal training, leave that class by design and are not targets of the critique. On *fixed location*, the gap $I(R;X)$ is defined for a fixed $\hat{\mu}$ and measures conditional residual shape, not the quality of the point forecast, as Remark 4 sets out. The identity is also *score-specific*. The exact mutual-information form belongs to the log-score, and for CRPS or the interval score the analogous gap is the regret of the best X-blind residual law under that score, which is not an information quantity. The density and log-score equalities of Propositions 2 and 3 are a large-calibration, or smoothed-density, statement, and what conformal delivers finite-sample is coverage rather than a density. Finally the figures and benchmark use synthetic generators with known ground truth, which is exactly what makes the oracle and $I(R;X)$ computable, and the identity itself is verified numerically by `check_gap.py`.

11 Conclusion

Conformal prediction certifies the coverage of a set, finite-sample and distribution-free, under exchangeability. That guarantee is genuine and, for the right objective, exact. The error is reading it as distributional forecast quality.

For a fixed location predictor the guarantee one can have, the marginal one, is rarely the guarantee one wants, and the conditional one is not available distribution-free without more structure. The output is a set and not a measure. Read as a single-shape distribution, its log-score loss against the oracle is exactly the residual-information term $I(R;X)$ of (4). Meanwhile the exchangeability that even the basic guarantee requires is usually missing in deployed time series.

Within the single-shape residual class conformal prediction is optimal, and where a guarantee is the product it is the right tool. But coverage answers the question *is the truth in this region?*, and in forecasting that is frequently not the question anyone was asking. This is not weak uncertainty quantification. It is exact uncertainty quantification for a set, a coverage guarantee, no more and no less. The gambit is permanent, the rent is $I(R;X)$, and one should at least know what one has paid.

Reproducibility. Every data figure is generated by the accompanying script `figures.py` (Figures 1–3, 4, 5, 7), and the time-series illustration by the benchmark scripts, all from the synthetic generators described in the text; Figure 6 is a schematic drawn by `fig_plane_schematic.py`. The identities of Proposition 3 are confirmed numerically by `check_gap.py`, which computes both sides independently (by grid integration) and matches them to machine precision on a heteroscedastic Gaussian and a skewed residual law. Code, figures, and an interactive companion are available at https://conformalprediction.net and https://github.com/microprediction/conformalprediction.

Data Availability Statement. This study uses no external or third-party data. All figures and numerical results are produced from the synthetic data-generating processes described in the text; the code that implements them and reproduces every figure, together with `check_gap.py` (the machine-precision numerical verification of Proposition 3), is openly available at https:

//github.com/microprediction/conformalprediction.

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